c-only slope law: observability matrix

1 Model

Gain law (slope only)

$$\bar{w} = \frac{w}{R}, \qquad \bar{w}_s = \text{visible surface}, \qquad h = \bar{w} - \bar{w}_s, \qquad \hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k], \qquad \hat{h}[k] = \hat{\bar{w}}[k] - \bar{w}_s$$

$$\bar{a}'(\hat{h};\, c) = \frac{1-c}{\hat{h}^2}, \qquad \frac{\partial \bar{a}'}{\partial c} = -\frac{1}{\hat{h}^2}, \qquad \frac{\partial \bar{a}'}{\partial \hat{h}} = -\frac{2\,\bar{a}'}{\hat{h}}$$

State equation (M = the step the estimator believes the particle makes)

$$\delta\bar{w}_1[k+1] = \delta\bar{w}_2[k]$$

$$\delta\bar{w}_2[k+1] = \delta\bar{w}_3[k]$$

$$\delta\bar{w}_3[k+1] = \lambda_c \cdot \delta\bar{w}_3[k] - F_{dw}[k] \cdot e_{\bar{a}_w}[k] - \varepsilon_w[k] \qquad M[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'\big(\hat{h}[k];\, c\big)\, M[k]$$

$$c[k+1] = c[k]$$

$$F_{dw}[k] \approx \big(1 + (1-\lambda_c)d\big)\,\frac{\Delta\bar{w}_d[k]}{\hat{\bar{a}}_w[k]} \qquad \text{(constant velocity)}$$

Output equation ($\nabla_d\bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$: the reading is the gain d steps ago)

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k], \qquad R_1 = \mathrm{Var}(n_w)$$

$$y_2[k] = \bar{a}_w[k-d] + n_a[k], \quad R_2 = \mathrm{Var}(n_a), \qquad \hat{y}_2[k] = \hat{\bar{a}}_w[k] - \bar{a}'\big(\hat{h}[k];\, \hat{c}[k]\big)\,\nabla_d\bar{w}_d[k]$$

Estimation error and output error

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{w}_1}[k] \\ e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_3}[k] \\ e_{\bar{a}_w}[k] \\ e_c[k] \end{bmatrix} = \begin{bmatrix} \delta\bar{w}_1 - \delta\hat{\bar{w}}_1 \\ \delta\bar{w}_2 - \delta\hat{\bar{w}}_2 \\ \delta\bar{w}_3 - \delta\hat{\bar{w}}_3 \\ \bar{a}_w - \hat{\bar{a}}_w \\ c - \hat{c} \end{bmatrix}, \qquad \begin{aligned} e_{y_1}[k] &= y_1[k] - \delta\hat{\bar{w}}_1[k] \\ e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] \end{aligned}$$

Error propagation (no noise, coefficients at the estimate)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k], \qquad \mathbf{e}[k_0+j] = \mathbf{\Phi}(k_0+j, k_0)\,\mathbf{e}[k_0], \qquad \mathbf{\Phi}(k_0+j, k_0) = \mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\ 0 & 0 & \bar{a}'(1-\lambda_c) + \dfrac{2\bar{a}'M[k]}{\hat{h}[k]} & 1 + \bar{a}'F_{dw}[k] & -\dfrac{M[k]}{\hat{h}[k]^2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} e_{y_1}[k] \\ e_{y_2}[k] \end{bmatrix} = \mathbf{H}[k]\,\mathbf{e}[k], \qquad \mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{2\bar{a}'\,\nabla_d\bar{w}_d[k]}{\hat{h}[k]} & 1 & \dfrac{\nabla_d\bar{w}_d[k]}{\hat{h}[k]^2} \end{bmatrix}$$

2 Observability matrix

Leading order used from here on: first order in $\bar{a}' F_{dw}$ and in $\bar{a}' \nabla_d \bar{w}_d / \hat{h}$

$$\beta_k := \frac{M[k]}{\hat{h}[k]^2}, \qquad \gamma_k := \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^2}, \qquad \mathbf{F}_e[k] \to \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\ 0 & 0 & 0 & 1 & -\beta_k \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H}[k] \to \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \gamma_k \end{bmatrix}$$

$$\text{dropped:} \quad F_{e,44} - 1 = \bar{a}' F_{dw}, \qquad F_{e,43} = \bar{a}'(1-\lambda_c) + \frac{2\bar{a}' M}{\hat{h}}, \qquad H_{23} = -\frac{2\bar{a}'\,\nabla_d \bar{w}_d}{\hat{h}}$$

$$\text{steady state of the } e_{\delta \bar{w}_3} \text{ loop:} \quad (F_{e,44} - 1) + F_{e,43}\Big(- \frac{F_{dw}}{1-\lambda_c}\Big) = -\frac{2\bar{a}' M F_{dw}}{\hat{h}\,(1-\lambda_c)}$$

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \mathbf{H}[k_0+j]\, \mathbf{\Phi}(k_0+j, k_0)\, \mathbf{e}[k_0], \qquad \beta_i := \beta_{k_0+i}, \;\; \gamma_j := \gamma_{k_0+j}, \;\; F_i := F_{dw}[k_0+i]$$

The matrix

$$\begin{bmatrix} e_{y_1}[k_0] \\ e_{y_2}[k_0] \\ e_{y_1}[k_0+1] \\ e_{y_2}[k_0+1] \\ e_{y_1}[k_0+2] \\ e_{y_2}[k_0+2] \\ e_{y_1}[k_0+3] \\ e_{y_2}[k_0+3] \\ e_{y_1}[k_0+4] \\ e_{y_2}[k_0+4] \\ \vdots \\ e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \gamma_0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\beta_0 + \gamma_1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1) + \gamma_2 \\ 0 & 0 & \lambda_c & -F_0 & 0 \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1+\beta_2) + \gamma_3 \\ 0 & 0 & \lambda_c^2 & -\lambda_c F_0 - F_1 & F_1\,\beta_0 \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1+\beta_2+\beta_3) + \gamma_4 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \lambda_c^{j-2} & -\sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_i & \sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_i \sum_{l=0}^{i-1} \beta_l \\ 0 & 0 & 0 & 1 & -\sum_{i=0}^{j-1} \beta_i + \gamma_j \end{bmatrix}}_{\mathbf{O}(k_0)} \begin{bmatrix} e_{\delta \bar{w}_1}[k_0] \\ e_{\delta \bar{w}_2}[k_0] \\ e_{\delta \bar{w}_3}[k_0] \\ e_{\bar{a}_w}[k_0] \\ e_c[k_0] \end{bmatrix}$$

The five columns (F_{dw}, β, γ constant)

	y_1 rows ($j = 0, 1, 2, \dots$)	y_2 rows ($j = 0, 1, 2, \dots$)
$e_{\delta \bar{w}_1}$	$1,\ 0,\ 0,\ 0,\ \dots$	$0,\ 0,\ 0,\ \dots$
$e_{\delta \bar{w}_2}$	$0,\ 1,\ 0,\ 0,\ \dots$	$0,\ 0,\ 0,\ \dots$
$e_{\delta \bar{w}_3}$	$0,\ 0,\ 1,\ \lambda_c,\ \lambda_c^2,\ \cdots \to 0$	$0,\ 0,\ 0,\ \dots$
$e_{\bar{a}_w}$	$0,\ 0,\ 0,\ -F_{dw},\ -F_{dw}(1+\lambda_c),\ \cdots \to -\dfrac{F_{dw}}{1-\lambda_c}$	$1,\ 1,\ 1,\ \dots$
e_c	$0,\ 0,\ 0,\ 0,\ F_{dw}\beta,\ F_{dw}\beta(2+\lambda_c),\ \cdots \approx \dfrac{F_{dw}}{1-\lambda_c}(j-3)\beta$	$\gamma,\ -\beta+\gamma,\ -2\beta+\gamma,\ \dots,\ -j\beta+\gamma$

3 What the matrix shows

$e_{\bar{a}_w}$ and e_c in the y_2 rows

$e_{\bar{a}_w}:\ 1,\ 1,\ 1,\ \ldots \qquad e_c:\ s_0,\ s_1,\ s_2,\ \ldots, \qquad s_j := -\sum_{i=0}^{j-1}\beta_i + \gamma_j$

$$\det\begin{bmatrix}1 & s_j\\ 1 & s_{j+1}\end{bmatrix} = s_{j+1} - s_j = -\beta_j$$

$$\bar{s} = \frac{\sum_j w_j\, s_j}{\sum_j w_j}, \qquad \sum_j w_j\,(s_j - \bar{s})^2, \qquad w_j = \frac{1}{R_2} + \frac{1}{R_1}\Big(\frac{F_{dw}}{1-\lambda_c}\Big)^2$$

Hold

$$\Delta\bar{w}_d = 0: \qquad \gamma_j = 0, \qquad M = (1-\lambda_c)\,\delta\hat{\bar{w}}_3, \qquad s_j = -\sum_{i<j}\beta_i \approx 0, \qquad s_j - \bar{s} \approx 0$$

Height

$$\beta = \frac{M}{\hat{h}^2}, \qquad \hat{h} = 22.2 \to 1.1: \quad \frac{1}{\hat{h}^2} = 0.002 \to 0.83$$

Constant velocity, $\Delta\bar{w}_d = (h_0 - h_b)\,T_s/T_d$

$$\sum_{i=0}^{j-1}\beta_i \approx -\int_{\hat{h}_0}^{\hat{h}_j}\frac{dh}{h^2} = -\Big(\frac{1}{\hat{h}_j} - \frac{1}{\hat{h}_0}\Big), \qquad s_j \approx \frac{1}{\hat{h}_j} - \frac{1}{\hat{h}_0}: \quad 22.2 \to 3\,R:\ 0.29, \qquad 22.2 \to 1.1\,R:\ 0.86$$

$$\gamma_j = \frac{d\,\Delta\bar{w}_d}{\hat{h}_j^2} \ \propto\ \frac{1}{T_d}, \qquad \frac{F_{dw}}{1-\lambda_c} \ \propto\ \frac{1}{T_d\,\hat{\bar{a}}_w}$$